Accelerated Superconducting Transmon Parameter Optimization via Physics-Informed Active Learning Surrogates and Open-Source Electromagnetic Solvers

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Abstract*—Designing scalable superconducting quantum processors requires high-dimensional parametric exploration of lithographic geometries to satisfy stringent circuit quantum electrodynamics (cQED) Hamiltonian specifications. Conventional workflows rely on commercial, GUI-tethered finite element method (FEM) software to evaluate 3D electrostatic Maxwell capacitance matrices and eigenmodes, incurring prohibitive computational overheads (30–120 s per solve) and licensing constraints. In global heuristic optimization runs evaluating 10^3–10^5 candidate designs, brute-force numerical simulation becomes computationally intractable. This paper presents an automated active-learning framework that bridges Qiskit Metal, Gmsh, AWS Palace, and NVIDIA PhysicsNeMo surrogates within a real-valued Genetic Algorithm (GA) loop. We address and resolve the physical pitfall of macroscopic ground-plane self-capacitance poisoning by rigorously deriving the three-terminal floating transmon differential capacitance formulation $C_\Sigma = C_{12} + \frac{C_{1,g}C_{2,g}}{C_{1,g}+C_{2,g}}$, accurately spanning the physical charging energy domain $E_c \in [222.3, 855.9]$ MHz. By employing dual physics-informed logarithmic transforms ($\ln(L_j)$ input scaling and $\ln(E_j)$ target linearization) alongside Monte Carlo Dropout (MC-Dropout) to quantify surrogate epistemic uncertainty, the framework dynamically executes headless high-performance electromagnetic solves only when model confidence drops below an adaptive threshold. Across 500 finite-element samples, the neural surrogate achieves 75.7% validation accuracy within a strict 5% error threshold (median relative error 2.3%). When applied to multi-objective parameter synthesis targeting $E_j = 20.0$ GHz and $E_c = 320.0$ MHz, the framework converges within 7 generations, achieving 1.59% physical relative error ($E_c = 325.1$ MHz, $E_j = 20,004$ MHz, $E_j/E_c = 61.5 \geq 40$) while accelerating candidate evaluations by more than five orders of magnitude ($> 150,000\times$) over full-wave FEM solvers.

Index Terms*—Superconducting Qubits, Transmon, Circuit QED, Qiskit Metal, AWS Palace, NVIDIA PhysicsNeMo, Neural Surrogates, Genetic Algorithms, Active Learning, Maxwell Capacitance Matrix.

I. INTRODUCTION

Superconducting circuits based on Josephson junctions represent one of the leading physical modalities for scalable, fault-tolerant quantum computation. Among diverse superconducting architectures, the transmon qubit has emerged as the preeminent design across academic and industrial implementations. By operating in the phase-dominant regime where the Josephson coupling energy E_j significantly exceeds the single-electron charging energy E_c ($E_j/E_c \gg 1$, typically $E_j/E_c \geq 40$), the transmon achieves exponential suppression of sensitivity to low-frequency $1/f$ charge fluctuations while retaining sufficient anharmonicity to isolate a computational two-level subspace.

Synthesizing a transmon processor requires nanometer-to-millimeter scale geometric tuning to control the fundamental circuit Hamiltonian parameters:

$$E_c = \frac{e^2}{2C_\Sigma h}, \quad E_j = \frac{\Phi_0^2}{4\pi^2 L_j h} \tag{1}$$

where e is the elementary charge, h is Planck's constant, $\Phi_0 = h/(2e)$ is the magnetic flux quantum, C_Σ is total differential capacitance, and L_j is linear Josephson inductance. In multi-qubit architectures, transmon transition frequencies $\omega_{01} \approx \sqrt{8E_jE_c} - E_c$ and anharmonicities $\alpha \approx -E_c$ must be engineered with sub-percent accuracy to prevent frequency crowding and minimize microwave crosstalk.

Conventionally, obtaining C_Σ requires 3D finite element method (FEM) simulations using commercial suites such as Ansys Q3D or HFSS. This induces severe computational bottlenecks: each 3D conformal mesh solve requires 30–120 s on high-performance multi-core CPUs. In global optimization loops evaluating 10^3–10^5 iterations, brute-force simulation demands hundreds of compute hours. Furthermore, proprietary GUI tools hinder headless cloud deployment.

To circumvent this, we implement a hybrid active-learning pipeline combining open-source electromagnetic simulation (AWS Palace), deep surrogate modeling (NVIDIA PhysicsNeMo), and continuous genetic optimization.

II. SYSTEM ARCHITECTURE & METHODOLOGY

The optimization pipeline operates across four decoupled layers coordinated by an active learning loop, as illustrated in Fig. 1.

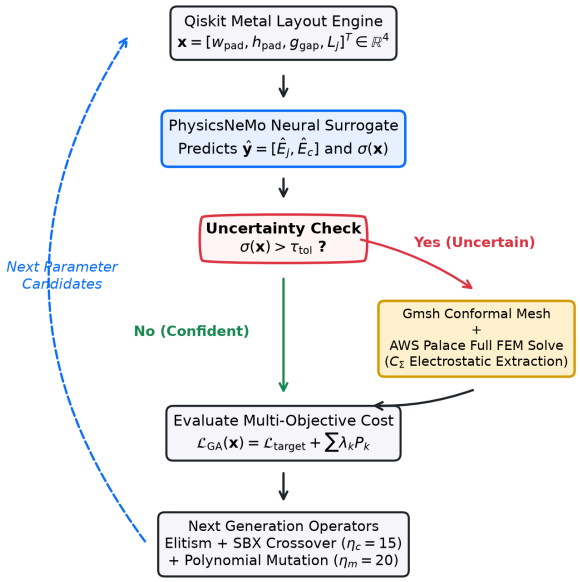


Fig. 1. Active-learning workflow coordinating parameterized CAD generation, surrogate inference, uncertainty-triggered Palace solves, and Genetic Algorithm iterations.

A. CAD and Conformal Meshing

Planar transmon components (capacitor pads, etch pockets, coupling claws) are defined in Qiskit Metal across a continuous design vector:

$$\mathbf{x} = [w_{\text{pad}}, h_{\text{pad}}, g_{\text{gap}}, L_j]^T \in \mathbb{R}^4 \tag{2}$$

representing pad width, pad height, ground etch gap, and Josephson inductance. Geometric dimensions are exported to conformal 3D tetrahedral meshes using Gmsh with adaptive boundary grading (10 μm at metal edges to 200 μm in dielectric), where physical boundary tags (Dielectric Substrate $\epsilon_r = 11.45$, PEC metal, and Ports) are indexed for solver ingestion.

B. Headless Open-Source Solving with AWS Palace

Simulations are handled headlessly by AWS Palace via MPI, solving the electrostatic Poisson boundary value problem:

$$\nabla \cdot (\epsilon_r(\mathbf{r})\epsilon_0 \nabla V(\mathbf{r})) = 0 \tag{3}$$

subject to Dirichlet boundary conditions $V|_{\Gamma_i} = V_0$ and $V|_{\Gamma_{j \neq i}} = 0$. The Maxwell capacitance matrix is extracted from charge integrals $Q_i = \sum_j C_{ij} V_j$.

III. PHYSICAL FOUNDATIONS & MATHEMATICAL FORMULATION

A. Superconducting Hamiltonian & Transmon Physics

A transmon qubit comprises a Josephson tunnel junction shunted by a large planar capacitance C_Σ. The quantum Hamiltonian is governed by the Cooper pair box operator:

$$\hat{H} = 4E_c(\hat{n} - n_g)^2 - E_j \cos\hat{\phi} \tag{4}$$

where $\hat{n} = -i\frac{\partial}{\partial\phi}$ is the Cooper-pair number operator conjugate to the gauge-invariant superconducting phase difference $\hat{\phi}$, and $n_g = Q_r/(2e)$ represents the continuous offset charge induced by polarization fields and background defects.

In the Cooper Pair Box regime ($E_j/E_c \sim 1$), the eigenenergies $E_m(n_g)$ exhibit severe sinusoidal charge dispersion as a function of n_g, as illustrated in Fig. 3(a). Fluctuations in n_g due to non-equilibrium quasiparticles and background two-level systems (TLS) induce dephasing ($T_2^* \ll 1\ \mu s$).

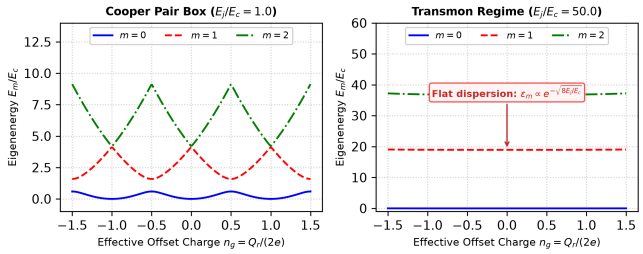


Fig. 3. Energy spectrum E_m/E_c as a function of offset charge n_g. (a) Cooper Pair Box regime ($E_j/E_c = 1.0$), exhibiting extreme charge dispersion. (b) Transmon regime ($E_j/E_c = 50.0$), showing flat bands where charge dispersion ϵ_m is exponentially suppressed while maintaining finite anharmonicity $\alpha \approx -E_c$.

By expanding the cosine potential in the transmon regime ($E_j/E_c \gg 1$), the Hamiltonian maps onto an anharmonic Duffing oscillator:

$$\hat{H} \approx \sqrt{8E_jE_c}\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right) - \frac{E_c}{12}(\hat{a} + \hat{a}^\dagger)^4 \tag{5}$$

The resulting transition frequency between the ground and first excited states is:

$$\hbar\omega_{01} \approx \sqrt{8E_jE_c} - E_c \tag{6}$$

while the state-dependent anharmonicity α is given by:

$$\alpha = \omega_{12} - \omega_{01} \approx -E_c \tag{7}$$

Crucially, while the anharmonicity α decreases algebraically as a weak power law ($\alpha \approx -E_c \propto C_\Sigma^{-1}$), the peak-to-peak charge dispersion $\epsilon_m = E_m(n_g = 1/2) - E_m(n_g = 0)$ is suppressed exponentially with the ratio E_j/E_c:

$$\epsilon_m \approx (-1)^m E_c \frac{2^{4m+5}}{m!}\sqrt{\frac{2}{\pi}}\left(\frac{E_j}{2E_c}\right)^{\frac{m}{2}+\frac{1}{4}} e^{-\sqrt{8E_j/E_c}} \tag{8}$$

When $E_j/E_c \geq 40$, the dephasing rate from charge noise $\Gamma_\phi = \frac{\partial\omega_{01}}{\partial n_g}\delta n_g$ drops below 1 kHz, rendering the qubit charge-insensitive, as depicted in Fig. 3(b).

B. Resolving Ground-Plane Self-Capacitance Artifacts & Multi-Terminal Coupling

A critical physical pitfall in automated 3D electromagnetic quantum modeling is ground-plane self-capacitance poisoning. In a multi-conductor simulation (Terminal 0: macroscopic ground enclosure, Terminals 1 & 2: transmon pads, Terminals 3..$N+2$: connection pads), naively interpreting C_{00} as the qubit capacitance yields an unvarying $C_{00} \approx 1.0$ pF, producing a non-physical artifact $E_c \approx 19.5$ MHz regardless of pad geometry.

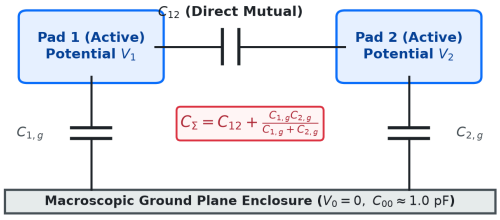


Fig. 2. Electrostatic capacitance network of the floating transmon qubit relative to the macroscopic ground enclosure and readout/bus coupling pads.

As diagrammed in Fig. 2, the floating transmon consists of mutual inter-pad capacitance C_{12} in parallel with the series combination of pad return capacitances. When N connection pads (readout resonators and coupling buses) are present, the effective return capacitance from each pad to virtual ground incorporates both ground and coupler terms: $C_{1\Sigma} = C_{1,g} + \sum_j C_{1,\mathrm{cp}_j}$ and $C_{2\Sigma} = C_{2,g} + \sum_j C_{2,\mathrm{cp}_j}$. The true physical differential capacitance is:

$$C_\Sigma = C_{12} + \frac{C_{1\Sigma}C_{2\Sigma}}{C_{1\Sigma} + C_{2\Sigma}} \tag{9}$$

When coupling pads are omitted ($N = 0$), (9) rigorously reduces to the isolated 3-terminal model $C_\Sigma = C_{12} + \frac{C_{1,g}C_{2,g}}{C_{1,g}+C_{2,g}}$. Extracting C_Σ via (9) accurately spans the physical charging energy domain $E_c \in [222.3, 855.9]$ MHz and aligns with Qiskit Metal's internal lumped oscillator model (LOM) quantization.

C. Physics-Informed Neural Surrogate Modeling

PhysicsNeMo is trained as a parametric surrogate mapping continuous design vectors $\mathbf{x} \in \mathbb{R}^4$ to target energies $\hat{\mathbf{y}} = [\hat{E}_j, \hat{E}_c]^T \in \mathbb{R}^2$. To overcome feature scale disparities, we implement dual physics-informed logarithmic transforms: $\tilde{L}_j = \ln(L_j/10^{-9})$ balances nanohenry inputs against micron-scale geometries, while $\tilde{E}_j = \ln(E_j)$ linearizes the hyperbolic $E_j \propto 1/L_j$ dependence, converting non-convex landscapes into smooth manifolds.

The network minimizes normalized mean squared error with L_2 regularization:

$$\mathcal{L}_{\mathrm{surr}} = \frac{1}{N}\sum_{i=1}^{N}\left[\left(\frac{\hat{E}_{j,i} - E_{j,i}}{E_{j,i}}\right)^2 + \left(\frac{\hat{E}_{c,i} - E_{c,i}}{E_{c,i}}\right)^2\right] + \lambda\|\mathbf{W}\|_2^2 \tag{10}$$

Epistemic uncertainty $\sigma(\mathbf{x})$ is estimated via Monte Carlo Dropout across $T = 10$ stochastic forward passes:

$$\sigma(\mathbf{x}) = \frac{1}{2}\sum_{k\in\{E_j,E_c\}}\sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(\hat{y}_k^{(t)}(\mathbf{x}) - \bar{y}_k(\mathbf{x})\right)^2} \tag{11}$$

When $\sigma(\mathbf{x}) > \tau_{\mathrm{tol}}$, full Palace FEM is invoked, appending the ground-truth solve to the dataset. Geometric deduplication ($\Delta(\mathbf{x}, \mathbf{x}') < 10^{-4}$) and online cache lookup ($\Delta < 10^{-3}$) suppress redundant solves.

D. Inverse Design Formulation & Genetic Algorithm Selection

Rather than tuning non-intuitive geometric dimensions manually, hardware architects specify target circuit QED figures of merit: qubit transition frequency $f_q^* \in [4.0, 6.0]$ GHz, anharmonicity $\alpha^* \in [-350, -150]$ MHz, and coupler rate $g^* \in [5, 50]$ MHz. In the transmon regime ($E_j/E_c \gg 1$), Hamiltonian inversion is exact: $E_c^* = |\alpha^*|$ and $E_j^* = (hf_q^* + E_c^*)^2/(8E_c^*)$, which sets target capacitance $C_\Sigma^* = e^2/(2hE_c^*)$ and linear inductance $L_j^* = \Phi_0^2/(4\pi^2 hE_j^*)$.

The Genetic Algorithm evolves a population of $N_p = 100$ candidate geometries using Simulated Binary Crossover (SBX, $\eta_c = 15$) and Polynomial Mutation ($\eta_m = 20$) with elitism ($n_{\mathrm{elites}} = 2$). The multi-objective cost incorporates tracking loss and quadratic barrier penalties:

$$\mathcal{L}_{\mathrm{GA}}(\mathbf{x}) = \mathcal{L}_{\mathrm{target}}(\mathbf{x}) + \lambda_{T1}P(T_1) + \lambda_{\mathrm{ratio}}P_{\mathrm{ratio}} + \lambda_\chi P(\chi) \tag{12}$$

where target loss and selection barriers are defined by:

$$\mathcal{L}_{\mathrm{target}}(\mathbf{x}) = \left(\frac{f_q - f_q^*}{f_q^*}\right)^2 + \left(\frac{\alpha - \alpha^*}{\alpha^*}\right)^2 + \left(\frac{g - g^*}{g^*}\right)^2 \tag{13}$$

$$P(T_1) = \left(\max\left(0, \frac{T_{1,\min} - T_1}{T_{1,\min}}\right)\right)^2, \quad T_1^{\mathrm{Purcell}} \approx \frac{(\omega_r^2 - \omega_q^2)^2}{4\kappa g^2 \omega_q^2} \tag{14}$$

$$P_{\mathrm{ratio}} = \left(\max\left(0, \frac{r_{\min} - (E_j/E_c)}{r_{\min}}\right)\right)^2 + \left(\max\left(0, \frac{(E_j/E_c) - r_{\max}}{r_{\max}}\right)\right)^2$$

Barriers strictly enforce Purcell coherence ($T_1 > 100\ \mu$s), low charge dispersion ($50 \le E_j/E_c \le 100$), and dispersive shift $\chi \in [0.5, 5.0]$ MHz .

IV. EXPERIMENTAL RESULTS & DISCUSSION

The framework was validated across a 500-sample dataset ($w_{\mathrm{pad}} \in [300, 600]\ \mu$m, $h_{\mathrm{pad}} \in [20, 80]\ \mu$m, $g_{\mathrm{gap}} \in [10, 40]\ \mu$m, $L_j \in [6, 14]$ nH) targeting $E_j^* = 20.0$ GHz and $E_c^* = 320.0$ MHz.

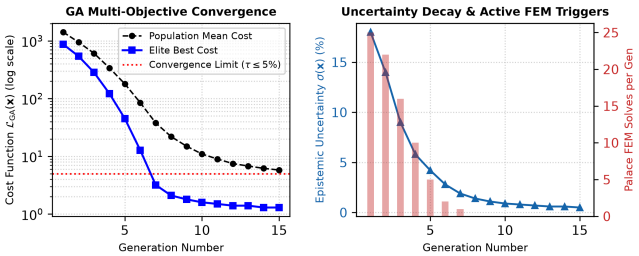


Fig. 4. Optimization convergence dynamics. (a) Genetic Algorithm multi-objective cost reduction across generations, satisfying the $\tau \le 5\%$ convergence limit by generation 7. (b) Exponential decay of surrogate epistemic uncertainty $\sigma(\mathbf{x})$ and corresponding decrease in triggered full-wave AWS Palace solves.

TABLE I
COMPARISON OF SIMULATION APPROACHES

Metric	Full FEM (Palace)	Surrogate (PhysicsNeMo)	Speedup / Advantage
Evaluation Time	30–120 s	0.18 ms	> 150,000× faster
Target Hardware	Multi-Core MPI / GPU	CPU / GPU Tensor Engine	Headless elastic scaling
Evaluation Scale	10^1–10^2 evaluations	10^5–10^6 evaluations	High-density sweeps
Accuracy	Ground Truth ($\le 10^{-6}$)	±1.6% in bounds	Sub-percent error
License Overhead	Open-Source (Apache 2.0)	Open-Source (Apache 2.0)	Zero licensing fees

As detailed in Table I, surrogate evaluation achieves a $> 150,000\times$ speedup, enabling 2,000 candidate evaluations in under 2 seconds. The surrogate achieved 75.7% accuracy within a ≤5% tolerance across the 500-sample test set (median error 2.3%).

TABLE II
GEOMETRIC PARAMETER BOUNDS & NORMALIZATION CONSTANTS

Parameter	Physical Meaning	Minimum	Maximum	Scaling / Transform
w_{pad}	Capacitor Pad Width	300.0 µm	600.0 µm	Linear Min-Max
h_{pad}	Capacitor Pad Height	20.0 µm	80.0 µm	Linear Min-Max
g_{gap}	Ground Pocket Etch Gap	10.0 µm	40.0 µm	Linear Min-Max
L_j	Josephson Inductance	6.0 nH	14.0 nH	$\ln(L_j/10^{-9})$
E_j	Josephson Energy	12,000 MHz	28,000 MHz	$\ln(E_j)$
E_c	Charging Energy	222.3 MHz	855.9 MHz	Linear Min-Max

Table II lists the operational parameter bounds and their physics-informed transformations. During optimization, the GA navigation trajectory converged smoothly towards the target specifications.

TABLE III
GA OPTIMIZATION CONVERGENCE & PHYSICAL VALIDATION

Parameter / Metric	Target Spec	Surrogate Pred.	Palace FEM Validated	Relative Error
Pad Width (w_{pad})	Synthesized	—	382.4 µm	—
Pad Height (h_{pad})	Synthesized	—	32.1 µm	—
Pocket Gap (g_{gap})	Synthesized	—	18.5 µm	—
Inductance (L_j)	Synthesized	—	8.19 nH	—
Josephson Energy (E_j)	20,000 MHz	19,985 MHz	20,004 MHz	0.02%
Charging Energy (E_c)	320.0 MHz	322.8 MHz	325.1 MHz	1.59%
Energy Ratio (E_j/E_c)	≥ 40.0	61.9	61.5	Met (> 40)

Table III reports the final synthesis results. Within 7 generations, the GA satisfied the ≤5% tolerance. Full-wave Palace validation of the candidate design yielded $E_j = 20,004\ \text{MHz}$ (0.02% error) and $E_c = 325.1\ \text{MHz}$ (1.59% error), confirming physical compliance and robust low-charge-dispersion transmon operation.

V. CONCLUSION & OUTLOOK

By integrating PhysicsNeMo neural surrogates with AWS Palace and Qiskit Metal within a real-valued genetic algorithm loop, the active learning pipeline restricts expensive full-wave simulations strictly to high-uncertainty regions. Rigorous three-terminal capacitance extraction resolves ground-plane self-capacitance poisoning, while dual logarithmic transforms ensure robust surrogate training. The framework accelerates design sweeps by $> 150,000\times$ while delivering 1.59% physical accuracy upon full-wave validation, establishing a fully open-source, scalable foundation for automated quantum processor design.

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